

1.

Problem Identification

The workstation is configured with IP Address 192.168.10.65, Subnet Mask 255.255.255.192 (126), and Default gateway 192.168.10.1. The system cannot communicate because the IP address is not in the required subnet 192.168.10.0/26.

Subnet Details.

Network Address = 192.168.10.64

Broadcast Address = 192.168.10.127

Host Range = 192.168.10.65 - 192.168.10.126

Usable Hosts: 62

The IP address 192.168.10.65 belongs to 192.168.10.64/26

Configuration Errors

- * Wrong IP address selected
- * Gateway is in another subnet
- * Devices are on different networks
- * Communication cannot occur.

Correct Configuration

Assign the following values:

- * IP address = 192.168.10.10
 - * Subnet Mask = 255.255.255.192 (126)
 - * Default Gateway = 192.168.10.1
- Now the workstation and gateway are in the same subnet.

RESULT:

- * Network Address = 192.168.10.0
- * Broadcast Address = 192.168.10.63
- * Host Range = 192.168.10.1 - 192.168.10.62
- * Usable Hosts = 62

Conclusion:

The problem was caused by an incorrect IP address. After assigning a valid IP address in

the 192.168.10.0/24 subnet, the workstation can communicate successfully with other devices in the network.

2.

Problem Identification

Two devices are configured with different IPv6 addresses but cannot communicate. The problem must be identified by checking the IPv6 addresses, network prefix, and subnet configuration.

IPv6 Address Details

Device A: 2001:DB8::FFFF:42:8329/64

Expanded: 2001:0DB8:0000:0000:0000:FFFF:0042:8329

Network Prefix: 2001:0DB8:0000:0000

Device B: 2001:DB8:0:0:1::1/64

Expanded: 2001:0DB8:0000:0000:0001:0000:0000:0001

Network Prefix: 2001:0DB8:0000:0000

Configuration Error

Different network prefixes
Devices are in different subnets
Direct communication fails.

Correct Configuration

Assign both devices to the same /64 subnet

Eg:

Device A: 2001:DB8::F100:42:E329/64

Device B: 2001:DB8::1/64

Now both devices belong to the same network and communication is possible.

Host Addresses

Prefix Length = 64

Host Bits = 64

Available Host Addresses = 2^{64}

Conclusion

The communication failure occurred because the devices had different network prefixes. After assigning both devices to the same /64 subnet, they can communicate successfully and the IPv6 network works correctly.

3.

Problem Identification

The Organization uses the network 192.168.1.0/24 with 126, 127, and 128 subnets. Some departments have too many IP addresses, while others have very few. This results in inefficient IP address allocation.

Subnet Analysis

- * 126 → 62 Usable hosts
- * 127 → 30 Usable hosts
- * 128 → 14 Usable hosts

Some departments do not use all the available IP addresses, causing unused address space.

Problems Identified

- * Unequal IP allocation
- * IP address wastage
- * Some departments lack addresses
- * Existing subnet plan is inefficient.

Optimization Using VLSM

- * Allocate larger subnet to larger departments.
- * Allocate smaller subnet to smaller departments.
- * Reduce unused IP addresses.
- * Improve address utilization.
- * Support future network growth.

Conclusion

Using VLSM (Variable Length Subnet Mask) provides better IP address allocation by assigning subnet sizes according to departmental needs. It reduces IP wastage, improves network efficiency, and supports future expansion without changing the existing network design.

Problem Identification

Two devices are connected to the same network but cannot communicate. The IPv6 addresses and network prefixes must be checked to identify the configuration problem.

IPv6 Address Analysis

- * Device A : 2001:DB8::FF00:42:8329/64
- * Network Prefix : 2001:DB8::0:0
- * Device B : 2001:DB8::0:0:1::1/64
- * Network Prefix : 2001:DB8::0:0:1
- * The devices have different network prefixes.

Configuration Errors

- * Different IPv6 network prefixes
- * Devices are in different subnets
- * Direct communication is not possible
- * Address configuration is incorrect.

Optimized Configuration

Assign both devices to the same /64 Subnet

Eg:

- * Device A : 2001:DB8::FF00:42:8329/64
- * Device B : 2001:DB8::1/64

Now both devices belong to the same network and communication is successful.

Conclusion:

The communication problem occurs because of the devices are configured in different IPv6 subnets. After assigning both devices to the same IPv6 network prefix, they can communicate correctly. The corrected subnet provides 2^{64} available host addresses, making it suitable for modern IPv6 networks.

Problem Identification

A router contains routing entries for different destination networks. User report that packets sent to 192.168.2.100 are not reaching the destination. The routing table must be checked to identify and correct the problem.

Routing Table Analysis

Destination: 192.168.1.0/24 → Next Hop 10.0.0.2

Destination: 192.168.2.0/24 → Next Hop 10.0.0.3

* Default Route $\rightarrow$ 10.0.0.1

* Destination IP: 192.168.2.100

* Matching Route: 192.168.2.0/24

* Next Hop: 10.0.0.3.

Configuration check

* Destination belongs to 192.168.2.0/24.

* Longest - prefix match is selected

* Packet is forwarded to 10.0.0.3.

* If communication fails, verify the routing entry and next-hop configuration.

Optimization

Verify routing table entries

Check next-hop IP address

Remove incorrect routes

Update routing information

Maintain correct default route.

Conclusion :

The packet for 192.168.2.100 matches the route 192.168.2.0/24 and is forwarded to 10.0.0.3. If no matching route exists, the router uses the default route (10.0.0.1). Correct routing tables ensure reliable and efficient packet forwarding.